

Real-Time Speech Emotion Recognition Using CNN-LSTM

1. Project Title

Real-Time Speech Emotion Recognition Using CNN-LSTM Deep Learning Architecture

2. Problem Statement

Human speech carries emotional information through variations in pitch, tone, energy, and speaking patterns. Traditional communication systems focus mainly on the spoken words and often ignore emotional cues.

The objective of this project is to develop a deep learning-based Speech Emotion Recognition (SER) system capable of identifying emotions from voice signals in real time using a hybrid CNN-LSTM architecture.

3. Objectives

1. Develop an intelligent system capable of recognizing emotions from speech.
 2. Extract meaningful audio features from speech signals.
 3. Build a CNN-LSTM model for emotion classification.
 4. Achieve high prediction accuracy on standard speech emotion datasets.
 5. Enable real-time emotion detection through microphone input.
 6. Visualize model predictions and performance metrics.
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4. Scope of the Project

The project focuses on recognizing emotions from speech audio using deep learning techniques.

Recognized emotions may include:

- Happy
- Sad
- Angry
- Fear
- Neutral
- Disgust
- Surprise

The system will process audio signals, extract features, classify emotions, and display predictions in real time.

5. Dataset

Dataset: RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song)

Dataset Features:

- Professional actor recordings
- Multiple emotional categories
- High-quality WAV files
- Widely used benchmark dataset

Input Format:

- WAV Audio Files

Output Labels:

- Emotion Categories
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6. System Architecture

Speech Input ↓ Audio Preprocessing ↓ Feature Extraction ↓ Mel Spectrogram Generation ↓ CNN Layers
↓ Feature Maps ↓ LSTM Layers ↓ Dense Layers ↓ Emotion Classification ↓ Real-Time Display

7. Methodology

Phase 1: Data Collection

- Download RAVDESS dataset
- Organize audio files
- Label emotions

Phase 2: Audio Preprocessing

- Load audio files using Librosa
- Normalize audio signals
- Remove unwanted noise
- Convert audio into Mel Spectrograms

Phase 3: Feature Extraction

Extract:

- Mel Spectrogram Features
- MFCC Features
- Spectral Contrast
- Chroma Features (optional)

Phase 4: CNN Feature Learning

CNN learns:

- Voice texture
- Pitch patterns
- Frequency structures
- Energy distribution

Phase 5: Temporal Learning using LSTM

LSTM learns:

- Sequential speech patterns
- Emotion transitions over time
- Long-term dependencies

Phase 6: Classification

Softmax layer predicts:

- Happy
- Sad
- Angry
- Fear
- Neutral
- Others

Phase 7: Real-Time Prediction

- Capture audio through microphone
 - Generate Mel Spectrogram
 - Feed into trained CNN-LSTM model
 - Display predicted emotion
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8. Technologies Used

Programming Language:

- Python

Libraries:

- TensorFlow
- Keras
- Librosa
- NumPy
- Pandas
- Matplotlib
- Scikit-Learn
- SoundDevice
- PyAudio

Development Environment:

- Jupyter Notebook
 - VS Code
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9. CNN-LSTM Architecture

Input Shape: 128×128 Mel Spectrogram

CNN Block:

- Conv2D
- ReLU
- MaxPooling

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- Conv2D
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Feature Flattening

LSTM Layer:

- 128 Units

Dense Layer:

- 64 Units

Output Layer:

- Softmax Activation

Emotion Classes

10. Evaluation Metrics

The model performance will be evaluated using:

- Accuracy
 - Precision
 - Recall
 - F1 Score
 - Confusion Matrix
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11. Expected Outcomes

- Accurate emotion recognition from speech
 - Real-time microphone-based emotion detection
 - CNN-LSTM deep learning implementation
 - Interactive emotion prediction system
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12. Future Enhancements

- Multilingual emotion recognition
 - Video + audio emotion detection
 - Transformer-based architectures
 - Deployment as a web application
 - Integration into virtual assistants
 - Mental health monitoring systems
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13. Project Timeline (7 Days)

Day 1:

- Study Speech Emotion Recognition

- Understand CNN and LSTM

Day 2:

- Dataset Collection
- Audio Preprocessing

Day 3:

- Mel Spectrogram Generation
- Data Preparation

Day 4:

- CNN Model Development

Day 5:

- CNN-LSTM Integration
- Model Training

Day 6:

- Evaluation and Testing

Day 7:

- Real-Time Microphone Integration
- Documentation and Presentation

14. Conclusion

This project presents a real-time Speech Emotion Recognition system using a hybrid CNN-LSTM architecture. The model leverages CNN layers for spatial feature extraction and LSTM layers for temporal sequence learning, enabling accurate recognition of emotions from speech signals. The developed system demonstrates practical applications in human-computer interaction, virtual assistants, customer service analytics, and healthcare monitoring.